

Step 1 – 3: Advanced Search & Methodology

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CS XI

Chapter 6: Digital Literacy

Lecture 11 of 14

★ Golden

Step 1 – 3: Advanced Search & Methodology ★ Golden

The **methodology** is the plan of methods a researcher uses to answer the question.

Step 1: Perform advanced searches

LEARN IT

Students first search for **background information** using search engines (such as Google or Bing). They use **advanced search operators** – special symbols or words typed into the search box – to **narrow down** results and find **reliable secondary data**.

Suggested query	Operator used	What it does
"mobile phone use" "study habits" students	" " (quotes)	Finds the exact phrases "mobile phone use" and "study habits", not the words scattered randomly.
"academic performance" filetype:pdf	filetype:	Finds only PDF documents (often official reports and research papers).
digital distraction site:.edu	site:	Searches only educational websites (addresses ending in .edu).

Other useful operators (background): - (minus) excludes a word, e.g. phone addiction -game; OR finds either word, e.g. smartphone OR mobile; site:.gov.pk searches only Pakistani government websites.

LEARN IT

Source evaluation checklist. Before trusting a source, ask:

- | | |
|---|--|
| ☐ Is the author/organisation clearly mentioned? | ☐ Is the information recent ? |
| ☐ Is the website reliable (.edu, .gov, research-based)? | ☐ Does the source provide facts/references , or is it biased (one-sided, trying to sell something)? |

WORKED EXAMPLE

Building a search query step by step. You want official PDF reports about screen time of students on Pakistani government websites.

Step 1 – Plain search: screen time students → millions of results, many blogs and adverts.

Step 2 – Exact phrase: "screen time" students → results must contain the exact phrase.

Step 3 – Only PDFs: "screen time" students filetype:pdf → mostly reports and papers.

Step 4 – Only government sites: "screen time" students filetype:pdf site:.gov.pk → a small number of highly relevant, reliable results.

Step 5 – Evaluate: open a result and check the checklist: author named? year recent? references given?

Step 2: Identify data types

LEARN IT

Students decide what kind of **primary data** they will collect:

- **Quantitative:** number of hours used daily; percentage who feel distracted.
- **Qualitative:** opinions on why they get distracted; suggestions for better use.

Step 3: Select data-collection methods

LEARN IT

Because they need to ask the **same questions to many students**, a **survey/questionnaire** (via Google Forms or on paper) is the **most efficient** method. Interviews would take too long for so many people, and observation could not reveal opinions.

REMEMBER IT

Three operators, three jobs: " " = **exact**, **filetype:** = **format**, **site:** = **where**. Note: there is **no space** after the colon (`site:.edu` , not `site: .edu`).

COMMON MISTAKE

Mistake: Assuming the first search result is the most reliable. Search results can be adverts or popular blogs. **Fix:** Always apply the source evaluation checklist. A website ending in .edu or .gov is usually more reliable, but you should still check the date and author.

اردو میں سمجھیں

پہلے قدم میں طلبہ سرچ انجن پر جدید سرچ آپریٹرز (Advanced Search Operators) استعمال کرتے ہیں: کوئیشن مارکس ("") بالکل وہی جملہ تلاش کرتے ہیں، filetype:pdf صرف پی ڈی ایف فائلیں اور site:.edu صرف تعلیمی ویب سائٹس دکھاتا ہے۔ ہر ذریعے کو مصنف، تاریخ، قابل اعتماد ویب سائٹ اور حوالوں کی بنیاد پر پرکھا جاتا ہے۔ پھر ڈیٹا کی اقسام طے کی جاتی ہیں اور بہت سے طلبہ سے سوال کرنے کے لیے سروے کو بہترین طریقہ چنا جاتا ہے۔

EXAM TIP

"Explain any three advanced search operators with examples" is a likely short question. For each, write the operator, an example query, and what it does. Also learn the four-point source evaluation checklist.

CHECK YOURSELF

1. Write a query that finds the exact phrase "online learning" only on educational websites.
2. Why was a survey chosen in Step 3?
3. Give one quantitative and one qualitative data item from Step 2.

Answers: 1. "online learning" site:.edu . 2. Because the same questions had to be asked to many students, and a survey is the most efficient way to do that. 3. Quantitative: hours used daily (or % who feel distracted). Qualitative: opinions on why they get distracted (or suggestions for better use).

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. The digital inquiry project applies:

- a. All of chapter 6 in order
- b. Only careers
- c. Only NAND
- d. Only OSI

MCQ 2. Advanced search helps you:

- a. Narrow secondary sources (quotes, site, date)
- b. Hack Wi-Fi
- c. Draw K-maps
- d. Sort in $O(1)$

MCQ 3. Primary collection in the project is often:

- a. A class survey
- b. A CPU design
- c. A TCP stack
- d. A waterfall contract

MCQ 4. You organise results in a:

- a. Spreadsheet + charts
- b. Random WhatsApp dump only
- c. NAND netlist
- d. Figma only

MCQ 5. The artefact is:

- a. The final digital product that answers the question
- b. The first brainstorm sticky
- c. The career list
- d. The teacher's attendance

MCQ 6. A conclusion should:

- a. Answer the question and admit limits
- b. Ignore the data
- c. Name and shame students
- d. Invent $n=1$ million

Extra short questions

1. List the project steps in order (grouped as in the book).
2. What is a digital artefact here?
3. Why include secondary if you surveyed?
4. One sentence of a good limitation.

Extra long question

1. Outline a 12-step inquiry: "Does extra screen time lower concentration in our class?" Name tools and the artefact.
★ Golden.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Search/method; survey/primary; secondary+sheet; analyse/conclude; artefact.
2. Slides, infographic, short video, site — that answers the inquiry question.
3. To compare your small sample with a published claim.
4. $n=40$ in one college, so we cannot speak for all Sindh.

Long

1. Steps 1–3 search; 4–6 survey; 7–9 PBS/article + sheet; 10–12 charts, conclusion, artefact (infographic+spreadsheet).
Ethics. One expected chart: hours vs self-reported focus.